

This Month's DVD Contents

This month's DVD accompanying EFY plus is a collection of a variety of free and open source electronics design and automation (EDA) tools for multiple platforms

SNEHA AMBASTHA

Fedora Electronic Lab (FEL)

FEL is a spin on Fedora that contains all the free and open source software that one would require for EDA. It provides all open source hardware design flows depending on the semiconductor industry's current trend. FEL as a platform has the ability to map the three methodologies—design, simulation and verification—that enable one to have a better hardware design flow and experience.

GENESIS

GENeral NEural Simulation System (GENESIS) is a software platform that allows users to simulate complex neural structures. The object-oriented programming approach used in this simulator allows the user to modify, reuse and exchange models or model components easily. It also enables designing with parallel computing, while working on large-scale neural projects.

Pcb

Pcb is a software tool for Windows for designing printed circuit boards. It has many different types of layouts, such as lines, arcs, polygons, elements (lines, pins and arcs), vias and text information, separated into different layers to enable a smooth design. It supports about eight different layers, in addition to the silk screen layer, and controls the visibility of surface mounted device (SMD) pads, pins and vias. It does not have any memory or any time-consuming

Popular resources

- > 7Zip. Version 9.20 (stable version, file archiver)
- > Java Runtime Environment. Version 8 update 25 (runs programs written in Java programming language)
- > VLC. Version 2.1.5 (media player)
- > Opera. Version 26.0.1656.60 (web browser)
- > Free Download Manager. Version 3.9.4 build 1482 (file download manager)
- > Thunderbird. Version 31.1.0 (e-mail application)
- > Mozilla Firefox. Version 34.0.5 (web browser)
- > Apache Open Office. Version 4.1.1 (open office)
- > Moon Secure AV (antivirus)

features like auto-router and auto-placement.

Greensocs 4.2.0

Greensocs is a free development kit used for producing systems on a chip using SystemC (a derivative of C), which is different from Verilog or very high speed integrated circuit (VHSIC) hardware description language (VHDL) that uses C. Although its main purpose is to encourage designers to use SystemC, it lowers the cost of infrastructure development, adoption and training. It allows EDA companies to build tools in support of the standards and increases the use of IP and interchange.

Octave

Octave is a free, open-source-interpreted programming language that enables numerical computations. Its syntax is quite similar to MATLAB and is made available under the GNU general-public licence (GPL). The latest version of Octave comes with a graphical user interface, which makes this tool more user-friendly. Although normally used through the interactive command-line interface, it can

also be used to write non-interactive programs.

SCADA ProcessViewer

SCADA ProcessViewer is a free and open source tool. It is a cross-platform SCADA tool written in free Pascal language on Lazarus integrated development environment (IDE) 0.9.28.2. The source code for this tool is also freely available and can be changed as per project requirements. The plus-point here is that the changed code can also be released to the open source community to help others working on similar projects. It is distributed in terms of GNU GPL version 2.

Virtual sound-level meter

A virtual sound-level meter is a MATLAB development for reading calibrated .wav files and analysing the field of sound with an actual sound-level meter. It analyses and implements many different types of waves from fast to slow, and from noise to spectrograms. The executable version of this tool does not require MATLAB, and it can work on Windows XP, Vista and Windows 7. ●

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